



Assignment 08: Modeling Interactions between Properties of Sound Systems

handed out: 26 June, 16:00
to be submitted by: 3 July, 22:00

In this task, you are going to work with lexical data from four language families that is taken from the North Eurasian lexicostatistical database (NorthEuraLex). The data comes in two TSV files: `languages.csv` contains relevant information about the languages in the sample (indexed by ISO codes), and `forms.csv` contains data about the realisations of hundreds of concepts across the languages, including derived data about length, longest consonant cluster length and (smoothed) vowel-consonant ratio for each of the forms.

Task 1: Data Preparation and Exploration

- Load both tables into Pandas dataframes. The information from the languages table will become your main dataframe that you will expand upon.
- Use the concept specific information from the forms table in order to enlarge your languages dataframe by the following columns (you can simply average across concepts):

- `avgLength`: average word length
- `avgCluster`: average length of longest consonant cluster per word
- `avgVowRatio`: average vowel-consonant ratio

- For each numerical variable in the database, find the languages with the minimum and maximum values.
- Visualise the pairwise joint distributions of all four numeric variables by generating a pairplot, using hue in order to distinguish the four language families (Indo-European, Uralic, Turkic, Nakh-Daghestanian).

Task 2: Testing whether Families Differ in Inventory Size

Let us ask our first research question: Do sound inventory sizes differ between language families, i.e. are there language families which systematically show larger sound inventory sizes than others?

- Find an adequate test, explaining how you went about making the decision.
- Find out which assumptions should hold to ensure the reliability of the test. If necessary, transform data so that the assumptions are met as well as possible. Comment on assumptions that might be violated.
- Perform the test using the `statsmodels` package, and interpret the results. Which answer to the initial question have you found?

Task 3: Attempting to Predict Average Word Length

Use the `statsmodels` package in order to fit linear regression models which try to predict the average word length based on inventory size, cluster length, and vowel ratio; compare all three single-predictor models with the three two-predictor models and the three-predictor model, and use some of the key measures of model quality to discuss which model is the most adequate. How much of the variation in average word length can your best model account for?

Task 4: Comparing Linear Regression Models

implement your best regression model from Task 3 using `scikit-learn`, and test whether any of the more advanced regression models (`Lasso`, `ElasticNet`) works better for the same predictors. You can do this without creating a training-test split, comparison in terms of R^2 scores will do (check the `scikit-learn` documentation to find out how to retrieve these scores). Try to interpret any unexpected result you might find.

Task 5: Can We Predict Phonemic Vowel Length?

In this final task, you will use `scikit-learn` in order to fit a logistic regression which predicts whether a language has a phonemic distinction in vowel length.

- a) Choose potential predictor variables among the ones for which you have data, perhaps with the help of the visualisation like the one from Task 1d).
- b) Fit the model. How well does it perform in terms of accuracy? Analyse the coefficients to find out which of the predictors you chose turn out to contribute the most.
- c) Add cross-validation to your logistic regression model in order to get an impression of its ability to generalise to unseen languages.